

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

**JEE(Main)****UNIT TEST****14-06-2020****JEE(Main+Advanced) : NURTURE COURSE [PHASE : S, I(A), I, II & III]**

Do not open this Test Booklet until you are asked to do so.

Read carefully the Instructions on this Test Booklet.

**PAPER-2**

Paper : Physics, Chemistry & Mathematics

**Important Instructions :**

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **75** questions. The maximum marks are **300**.
5. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **25 questions** in each subject and each subject having **Two sections**.
  - (i) Section-I contains 20 multiple choice questions with **only one correct** option.  
**Marking scheme** : +4 for correct answer, 0 if not attempted and -1 in all other cases.
  - (ii) Section-II contains 5 **Numerical Value Type** questions  
**Marking scheme** : +4 for correct answer and 0 in all other cases.
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited**.
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

Name of the Candidate (in Capitals) \_\_\_\_\_

Form Number : in figures \_\_\_\_\_

: in words \_\_\_\_\_

Centre of Examination (in Capitals) : \_\_\_\_\_

Candidate's Signature : \_\_\_\_\_ Invigilator's Signature : \_\_\_\_\_

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Corporate Office : **ALLEN** CAREER INSTITUTE, "SANKALP", CP-6, Indra Vihar, Kota (Rajasthan) INDIA-324005

☎ +91-744-2757575 ✉ info@allen.ac.in 🌐 www.allen.ac.in

**ALLEN**  
**PART 1 - PHYSICS**

**SECTION-I : (Maximum Marks : 80)**

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

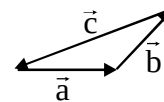
*Full Marks* : +4 If only the bubble corresponding to the correct option is darkened.

*Zero Marks* : 0 If none of the bubbles is darkened.

*Negative Marks* : -1 In all other cases

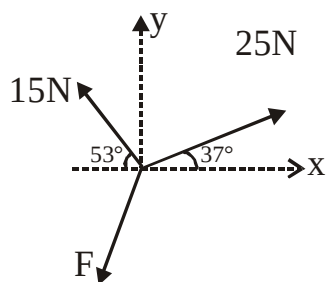
1. A ball is released from some height & rebounds with same speed after hitting the ground. If it travels 20 m in first 8 seconds find the no of collision made.
- (A) 8 (B) 9  
(C) 6 (D) 4

2. A little masked girl is holding a helium-filled balloon with a string while riding in a closed elevator going down a very tall building at constant speed. There is vacuum in elevator. Suddenly the elevator cable snaps, sending the elevator into free fall. Being shocked, the girl lets go of the string. She is even more surprised to see :-  
(A) the balloon rising  
(B) the balloon floating downward  
(C) the balloon remaining stationary  
(D) the balloon bouncing slowly between the floor and the ceiling
3. At a distance 0.6 km from the traffic light, breaks are applied to a car moving at a velocity  $15 \text{ ms}^{-1}$ . Determine the position of car relative to the traffic light 1.25 minute after the application of breaks if its retardation is  $0.3 \text{ ms}^{-2}$ .  
(A) 25 m (B) 43.75 m  
(C) 225 m (D) 375 m
4. Mark incorrect statement(s) for  $\vec{a}$ ,  $\vec{b}$  and  $\vec{c}$  shown in above diagram :

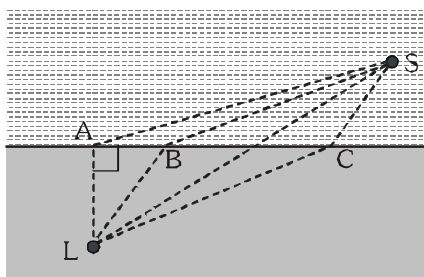


- (A)  $-\vec{a} - \vec{b} = \vec{c}$  (B)  $\vec{a} + \vec{c} = -\vec{b}$   
(C)  $\vec{a} + \vec{c} = \vec{b}$  (D)  $\vec{b} + \vec{c} = -\vec{a}$

5. Three forces are acting on an object shown in diagram. Their resultant is zero. The  $\vec{F}$  is :-

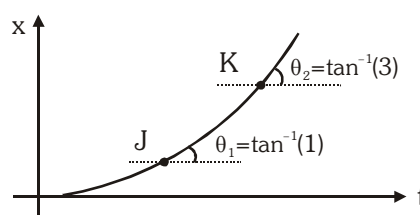


- (A)  $-11\hat{i} - 27\hat{j}$  (B)  $-20\hat{i} - 27\hat{j}$   
 (C)  $11\hat{i} - 3\hat{j}$  (D)  $20\hat{i} - 3\hat{j}$
6. You are a lifeguard at L on the sand and a swimmer S gets into difficulty in the ocean. Which path would take the shortest time for you to reach the swimmer, if you can walk faster than you can swim?



- (A) Path LAS.  
 (B) Path LBS.  
 (C) Straight line path LS.  
 (D) Path LCS.

7. A stone falls freely from a point A. It passes through the points  $B_1, B_2, B_3, \dots$  such that  $AB_1, AB_2, AB_3, \dots$  are in geometric progression. Then magnitude of velocities of stone at  $B_1, B_2, B_3, \dots$  are in  
 (A) arithmetic progression  
 (B) geometric progression  
 (C) harmonic progression  
 (D) none of these
8.  $x-t$  graph for a uniformly accelerated motion is as shown in the figure. Then find the average velocity between points J and K



- (A) 3m/s (B) 2m/s  
 (C) 4 m/s (D) 1.5 m/s
9. Given that force =  $\frac{\alpha}{\text{density} + \beta^3}$ , what are the dimensions of  $\alpha$  and  $\beta$  ?  
 (A)  $ML^{-2}T^{-2}$ ,  $ML^{-1/3}$  (B)  $M^2L^4T^{-2}$ ,  $M^{1/3}L^{-1}$   
 (C)  $M^2L^{-2}T^{-2}$ ,  $M^{1/3}L^{-1}$  (D)  $M^2L^{-2}T^{-2}$ ,  $ML^{-3}$

10. In a book, the answer for a particular question is expressed as

$$b = \frac{ma}{k} \log \left[ 1 + \frac{kl}{ma} \right]$$

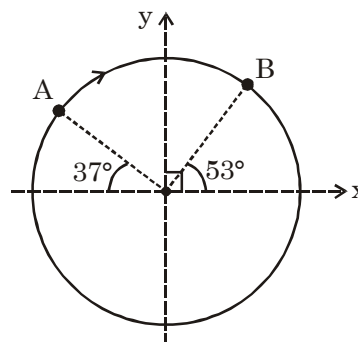
here  $m$  represents mass,  $a$  represents accelerations,  $\ell$  represents length. The dimension formula of  $b$  should be

- (A)  $M^1T^{-1}$  (B)  $M^0L^1T^0$   
 (C)  $M^0L^1T^{-1}$  (D)  $M^1L^2T^{-2}$
11. In a given system of units, 1 unit of mass = 2kg, 1 unit of length = 5m and 1 unit of time = 15sec. Then in this system, 12 Newton is equal to X units of force. Then X is
- (A) 270 (B) 12  
 (C) 240 (D) 300
12. x-coordinate of a particle moving along x-axis is  $x = \frac{t^3}{3} + t^2 - 3t$ . Here  $x$  is in metre and  $t$  in second. The motion starts at  $t = 0$ . Find acceleration of particle (in  $m/s^2$ ) when particle comes to instantaneous rest.
- (A) 2 (B) 4  
 (C) 5 (D) 3

13.  $\hat{a} + \hat{b} + \hat{c} = 0$  if  $\hat{a}, \hat{b}, \hat{c}$  are unit vectors and

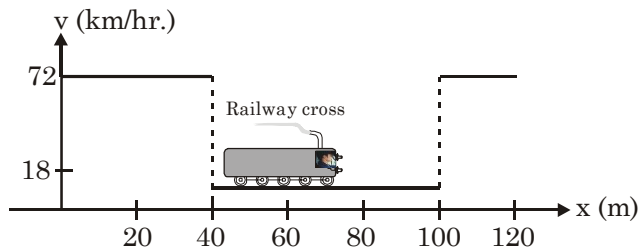
$\hat{a} \cdot \hat{b} - \hat{b} \cdot \hat{a} - \hat{c} \cdot \hat{a} = x$ . Then  $4x$  is equal to.

- (A) 4 (B) 6  
 (C) 3 (D) 2
14. A particle starts from A with constant speed  $v$  on the circle of radius  $R$ . Find the magnitude of average velocity from A to B:

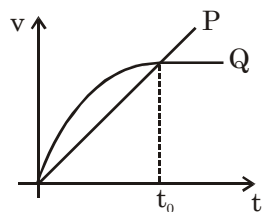


- (A)  $\frac{2v}{\pi}$  (B)  $\frac{2\sqrt{2}v}{3\pi}$   
 (C)  $\frac{2\sqrt{2}v}{\pi}$  (D)  $\frac{v}{\pi R}$

15. A car is travelling on a straight highway with a uniform speed of 72 km/hr. Due to a traffic jam near just opened Railway crossing it slowed down its speed to 18 km/hr as shown in v-x graph. The time (in sec) lost in crossing this traffic is (time taken for sudden change of speed is neglected) :-



- (A) 4 (B) 3 (C) 2 (D) 9
16. The velocity time graph for two particles P and Q moving in x-direction starting from same point is shown in figure. Choose the **CORRECT** alternative.

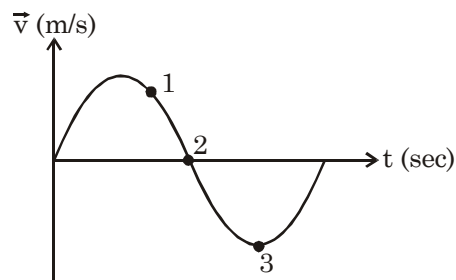


- (A) Initially particle Q leads over particle P  
 (B) At time  $t_0$  particle P overtakes particle Q.  
 (C) At time  $t_0$  particle Q overtakes particle P.  
 (D) There is no instant where the distance between the two neither increasing nor decreasing

17. Position of a particle is given by  $x = t(t - 1)(t - 2)$  where 't' is time then total number of turning points of particle after  $t = 0$  is :-

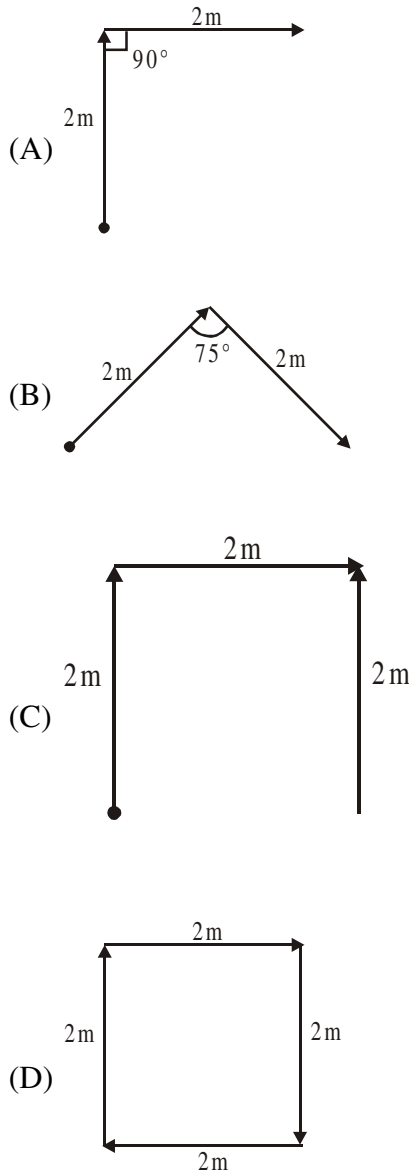
(A) 1 (B) 2 (C) 3 (D) 4

18. Figure shows the velocity-time (v-t) graph of a car travelling on a straight road. Which of the following statement is **CORRECT**?



- (A) Velocity and acceleration have same sign at point 1.  
 (B) Particle changes its direction of motion at point 3.  
 (C) Velocity of car is equal to zero but acceleration is not equal to zero at point 2.  
 (D) Speed of particle is continuously decreasing from point 1 to 3.

19. Which vector diagram represents the greatest magnitude of net displacement for an object ?



20. The horizontal and vertical displacements of a projectile at time 't' are  $x = 36t$  and  $y = 27t - 4.9t^2$  respectively where x and y are in meter and t is in second. Initial speed of the projectile in m/s is:  
(A) 15 (B) 30 (C) 45 (D) 60

**SECTION-II : (Maximum Marks: 20)**

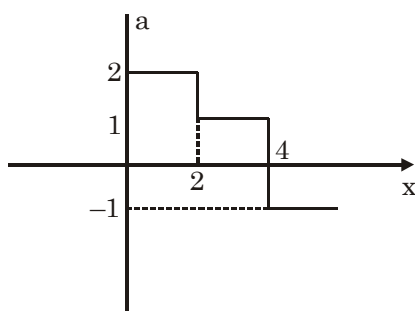
- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

**For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

| + |   |   |   |   |   |   |   |   |   | - |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 | 9 |

- Answer to each question will be evaluated according to the following marking scheme:  
*Full Marks* : +4 If **ONLY** the correct numerical value is entered as answer.  
*Zero Marks* : 0 In all other cases.

1. A cyclist is moving at 2 m/s at a distance 199 m from his Biker friend. Now Biker initially at rest wants to catch cyclist at his speed. The minimum time Biker would take if he can have a max acceleration/retardation of  $2 \text{ m/s}^2$ .
2. A particle starts from rest whose a-x graph is as shown. Find total distance travelled before it comes to rest.



3. A ball is thrown upwards from a tower takes 9 sec to hit ground. Another ball if thrown downwards with same speed takes 4 sec to hit the ground. If it is just released from tower than time taken (in sec) to hit the ground will be.
4. An insect moving along a straight line, has a displacement (in m) in every second equals to the magnitude of time elapsed. Assuming acceleration to be constant, then the magnitude of initial velocity of insect is  $10\alpha \text{ cm/s}$  then the value  $\alpha$  is :-
5. A rope of length 1.8 m is hanging from a hook. It is released and then it falls freely. If the time (in sec) taken by the rope to cross a point 5 m below the hook is x value of  $10x$  is :

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**PART 2 - CHEMISTRY**

**SECTION-I : (Maximum Marks : 80)**

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :  
*Full Marks* : +4 If only the bubble corresponding to the correct option is darkened.  
*Zero Marks* : 0 If none of the bubbles is darkened.  
*Negative Marks* : -1 In all other cases

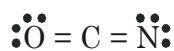
1. Which among the following options has greatest value?  
 (A)  $IE_1(\text{Na}^+)$  (B)  $IE_1(\text{F}^-)$   
 (C)  $IE_1(\text{O}^-)$  (D)  $IE_1(\text{Cl}^-)$
2. Consider the following two hydroxides :  
 XOH and YOH. (X and Y are any two elements)  
 Given :  
 $IE(\text{X}^-) = 4\text{eV/atom}$ ;  $E.A.(\text{X}^+) = 13\text{eV/atom}$   
 $IE(\text{Y}^-) = 0.6\text{eV/atom}$ ;  $E.A.(\text{Y}^+) = 5\text{eV/atom}$   
 Carefully read the following statements?  
 (I) YOH is acidic, whereas XOH is basic  
 (II) In X-O-H; the X-O bond is more polar compared to O-H bond

(III) If YOH reacts with HCl (Hydrochloric acid) then no reaction occurs.

Pick the **INCORRECT** statements

- (A) I, II only  
 (B) III only  
 (C) II only  
 (D) I, II, III

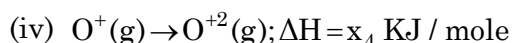
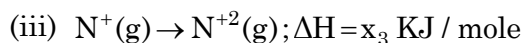
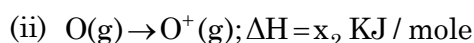
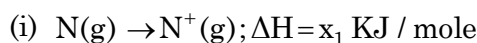
3. In the Lewis dot structure of :



the formal charges on N, C and O are respectively :

- (A) -1, 0, +1 (B) -1, 0, -1  
 (C) +1, 0, -1 (D) -1, 0, 0

4. Given below are four reactions with their enthalpies :



If it is known that  $x_1, x_2, x_3, x_4$  have their values from :

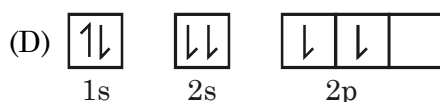
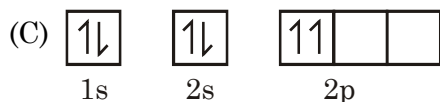
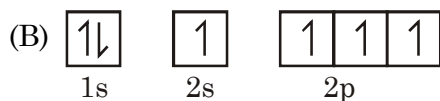
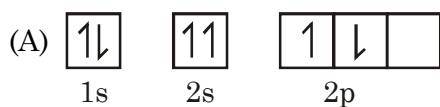
3400, 1310, 2856, 1400 (not in order);

then find the **INCORRECT** option

- (A)  $x_3 = 2856$   
 (B)  $x_2 = 1310$   
 (C)  $x_1 = 1400$   
 (D)  $x_3 = 3400$

5. Select the **INCORRECT** option  
 (A)  $I.E.(F) + E.A.(F) > I.E.(Cl) + E.A.(Cl)$   
 (B)  $I.E.(F) + I.E.(Cl) > E.A.(F) + E.A.(Cl)$   
 (C)  $I.E.(F) - I.E.(Cl) < E.A.(Cl) - E.A.(F)$   
 (D)  $I.E.(F) - I.E.(O) > E.A.(O) - E.A.(F)$
6. Find the total number of possible sets of four quantum numbers for the  $89^{th}$  electron of Ac ( $Z = 89$ )  
 (A) 5 (B) 14  
 (C) 10 (D) 6
7. Suppose more elements are discovered and  $8^{th}$  period is also present in Modern periodic table, then the inert gas of the period will be :  
 (A) Upn  
 (B) Uho  
 (C) Uth  
 (D) None of these
8. Pick the element with least metallic character among. K, Mg, Al and B  
 (A) K (B) Mg  
 (C) Al (D) B
9. The second electron gain enthalpy ( $EGE_2$  or  $\Delta H_{eg2}$ ) for oxygen and sulphur are respectively (in KJ/mole)  
 (A)  $-780, +590$   
 (B)  $-590, +780$   
 (C)  $-590, -780$   
 (D)  $+780, +590$
10. In the electronic configuration of Cl, find minimum electrons for which :  
 $n + \ell + m = 3$   
 (A) 6 (B) 4  
 (C) 5 (D) 8
11. Which of the following is **INCORRECT**?  
 (A)  $Na^+(aq) < Cs^+(aq)$  {Ionic mobility}  
 (B)  $S > O < Se$  (E.A.)  
 (C)  $Tl > Ga$  ( $I.E_1$ )  
 (D)  $3p < 3d$  {Energy in  $Li^{+2}$ }
12. A representative element : X has following values of successive I.E. (in KJ/mole)
- |        |        |        |        |        |
|--------|--------|--------|--------|--------|
| $IE_1$ | $IE_2$ | $IE_3$ | $IE_4$ | $IE_5$ |
| 590    | 1150   | 4920   | 6400   | 8100   |
- What can be the formula of oxide of this element.  
 (A)  $X_2O$  (B)  $XO_2$   
 (C)  $XO$  (D)  $X_2O_3$
13. Consider the following species :  
 (i)  $Na^+$  (ii)  $Mn^{+2}$   
 (iii)  $Cr^{+3}$  (iv)  $Cu^+$   
 Which are expected to be weakly repelled in external magnetic field?  
 (A) (i), (ii)  
 (B) (ii), (iii)  
 (C) (iii)  
 (D) (i), (iv)

14. In which of the following configuration only Pauli's rule is violated?



15. If we move on Z-axis, the probability of finding electron is zero or minimum in following orbitals except for?



16. Which of the following is not a drawback of Mendeleev's periodic table ?

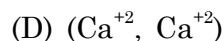
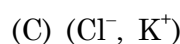
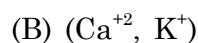
(A) There were some pair of elements which did not follow the increasing order of atomic weights.

(B) There were some groups in which very dissimilar elements were placed.

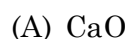
(C) Blank spaces were left for undiscovered elements

(D) There was no place for isotopes.

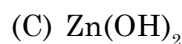
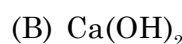
17. Out of  $\text{Cl}^-$ ,  $\text{Ca}^{+2}$  and  $\text{K}^+$  the pair of species showing maximum  $\Delta H_{\text{I.E.}}$  and minimum size respectively



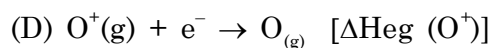
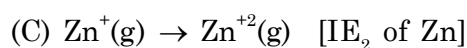
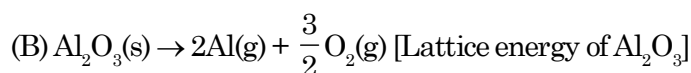
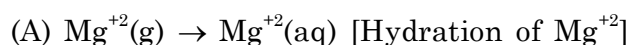
18. Which among the following has greatest lattice energy?



19. Which of the following can react with  $\text{KOH(aq.)}$  ?



20. Which of the following equation is **INCORRECTLY** matched?



SECTION-II : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

**For Example :** If answer is -77.25, 5.2 then fill the bubbles as follows.

| + |   | - |   |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 |
| 2 | 2 | 2 | 2 |
| 3 | 3 | 3 | 3 |
| 4 | 4 | 4 | 4 |
| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

- Answer to each question will be evaluated according to the following marking scheme:  
**Full Marks :** +4 If **ONLY** the correct numerical value is entered as answer.  
**Zero Marks :** 0 In all other cases.

- Find total number of lone pairs in  $\text{COF}_2$
- Scandium (Sc : Z = 21) and Yttrium (Y : Z = 39) are elements of lanthanide series. If according to Slater's rule the value of  $Z_{\text{eff}}$  of Sc is 'a' and that of Y is 'b', then find : (a - b)?
- Find spin magnetic moment ( $\mu_{\text{spin only}}$ ) of  $\text{Ni}^{+2}$  ion in B.M( Bohr magneton)
- Consider the following list of oxides  
 $\text{CO}$ ,  $\text{Li}_2\text{O}$ ,  $\text{CO}_2$ ,  $\text{Cs}_2\text{O}$ ,  $\text{MgO}$ ,  $\text{Al}_2\text{O}_3$ ,  $\text{BeO}$ ,  $\text{SO}_2$ ,  $\text{N}_2\text{O}_5$   
 Number of amphoteric oxides = x  
 Number of only acidic oxides = y  
 Find  $\left(\frac{y}{x}\right)$ ?
- Find bond length (in pm) of : (F - Cl) bond  
 Given : Bond length of  $\text{F}_2$  = 144 pm  
 Bond length of  $\text{Cl}_2$  = 199 pm  

$$\begin{cases} \text{E.N.}(\text{Cl}) = 3 \\ \text{E.N.}(\text{F}) = 4 \end{cases}$$

**ALLEN**  
**PART 3 - MATHEMATICS**

**SECTION-I : (Maximum Marks : 80)**

- This section contains **TWENTY** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :  
*Full Marks* : +4 If only the bubble corresponding to the correct option is darkened.  
*Zero Marks* : 0 If none of the bubbles is darkened.  
*Negative Marks* : -1 In all other cases

1. For the series 2, 5, 8, 11, ... up to  $n$  terms,  $T_n$  &  $S_n$  are  $n^{\text{th}}$  term & sum of  $n$  terms of given series respectively. Then value of  $T_{11} + S_{11}$  is :-  
 (A) 107 (B) 119  
 (C) 222 (D) 219
2. Value of  $\log_9 \sqrt[4]{\frac{729}{3^3 \times 9 \times 81}}$  :-  
 (A)  $\frac{1}{2}$  (B) 1  
 (C)  $\frac{3}{2}$  (D) 2

3. Sum of all possible values of  $x$  satisfying the equation  $\sqrt{x+2} = x$ , is \_\_\_\_\_  
 (A) 1 (B) 2  
 (C) 3 (D) 4
4. Four number are in increasing A.P. If their sum is 40 and the sum of their squares is 480 then the middle terms are :-  
 (A) 2, 6 (B) 4, 8  
 (C) 6, 10 (D) 8, 12
5. If  $2x^{\log_4 3} + 3^{\log_4 x} = 27$ , then sum of possible values of  $\sqrt{x}$  is equal to :-  
 (A) 0 (B) 2  
 (C) 4 (D) 6
6. Least positive integral value of  $x$  such that  $\frac{x^2 + 4x + 4}{2x^2 - x - 1} > 0$ , is :-  
 (A) 1 (B) 2  
 (C) 3 (D) 4
7. If value of  $(49)^{(1-\log_7 2)} + (5)^{-\log_5 4}$  is  
 (A)  $\frac{5}{2}$  (B)  $\frac{15}{2}$  (C)  $\frac{25}{2}$  (D)  $\frac{35}{2}$
8. Value of  $x$  satisfying the equation  $\log_2(9 - 2^x) = x - 3$ , is :-  
 (A) 0 (B) 1  
 (C) 2 (D) 3

9. If  $\alpha, \beta, \gamma$  are roots of the cubic equation  $x^3 + 2x - 1 = 0$  then value of  $(\alpha + \beta)(\beta + \gamma)(\gamma + \alpha)$  is :-  
 (A) 1 (B) -1  
 (C) 2 (D) -2
10. If  $\alpha$  &  $\beta$  are roots of the equation  $x^2 - 3x + 1 = 0$  then value of  $\frac{1}{\alpha} + \alpha^2 + \frac{1}{\beta} + \beta^2$  is :-  
 (A) 5 (B) 10  
 (C) 15 (D) 20
11. If  $4x - 1 = \sqrt{5}$ , then the value of  $4x^2 - 2x + 1$  is :-  
 (A) 4 (B) -4  
 (C) 2 (D) -2
12. If  $x = 2 + 2^{\frac{1}{3}} + 2^{\frac{2}{3}}$  then value of  $(x^3 - 6x^2 + 6x)$  is :-  
 (A) 12 (B) 8  
 (C) 2 (D) 14
13. If  $(a + b)(b + c)(c + a) = 11$  &  $a + b + c = 11$  then value of  $a^3 + b^3 + c^3$  is :-  
 (A) 1320 (B) 1364  
 (C) 1164 (D) 1298
14. If quadratic equations  $x^2 + px + q = 0$  &  $2x^2 - 4x + 5 = 0$  have a common root then  $p + q$  is never less than  $\{p, q \in \mathbb{R}\}$  :-  
 (A) 0 (B) 1  
 (C) 2 (D) 3
15. The product of all the solutions of the equation  $x^{1+\log_{10} x} = 1000x$  is :-  
 (A)  $10^{\sqrt{3}}$  (B)  $10^3$   
 (C)  $10^{-3}$  (D) 1
16.  $y = x^2 - 6x + 15$ ,  $x \in [-2, 3]$  then least value of  $y$  is :-  
 (A) 0 (B) 6  
 (C) 10 (D) 31
17. If  $\log_5 \log_4 \log_3 \log_2 (N) = 0$ , then number of digits in  $N \times 2^{19}$  are -  
 {where  $\log_{10} 2 = 0.3010$ } :-  
 (A) 21 (B) 31  
 (C) 20 (D) 30
18. Number of integral values of  $m$  for which roots of quadratic equation  $(m - 1)x^2 + 3x + (m - 3) = 0$  lie either side of origin, is/are :-  
 (A) 1 (B) 2  
 (C) 3 (D) 4
19. Product of all the integral values, which lie in the range of the expression  $\frac{16x^2 - 12x + 9}{16x^2 + 12x + 9}$ ;  $\forall x \in \mathbb{R}$ , is :-  
 (A) 2 (B) 4  
 (C) 6 (D) 8
20.  $|x - 1| + 2 \leq \log_2 \sqrt[3]{(7 + \sqrt{625})^3}$   
 number of integral solutions are :-  
 (A) 0 (B) 5  
 (C) 7 (D) 10

SECTION-II : (Maximum Marks: 20)

- This section contains **FIVE** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

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| 5 | 5 | 5 | 5 |
| 6 | 6 | 6 | 6 |
| 7 | 7 | 7 | 7 |
| 8 | 8 | 8 | 8 |
| 9 | 9 | 9 | 9 |

- Answer to each question will be evaluated according to the following marking scheme:  
**Full Marks :** +4 If **ONLY** the correct numerical value is entered as answer.  
**Zero Marks :** 0 In all other cases.

1. If quadratic equation

$$x^2 - (k + 4)x + \left(\frac{3k^2}{4} + 6\right) = 0 \text{ have equal roots,}$$

then the value of  $\left(k + \frac{3}{2}\right)$  is :-

2. Sum of all the solution(s) of the equation  $\log_{10}(x) + \log_{10}(x + 2) - \log_{10}(5x + 4) = 0$  is
3. If  $A = \log_3[4\cos^2(x) + 4\sin^2(x) - 1]$

$$B = \log_{\sqrt{2}-1}(\sqrt{2} + 1)$$

$$C = (2)^{\log_6 18} \cdot (3)^{\log_6 3}$$

then value of  $\frac{A}{B+C}$  is :-

4. Value of  $x$  satisfying the equation  $\sqrt{x^2 + 5x - 14} + |x^2 + 4x - 12| = 0$  is
5. If  $\alpha, \beta$  &  $\gamma$  are 3 values of  $x$   $\{\alpha > \beta > \gamma\}$  satisfying the equation  $(x)^{\log_2^2(x) + 4\log_2(x) + 3} = x^{1 + \log_2 x}$  then the value of  $(\alpha + \beta - \gamma)$  is :-

